Questions

Question 1 **2**

Question 2 **4**

Question 3 **6**

Question 4 **8**

Question 5 **10**

Question 6 **12**

Question 7 **14**

Question 8 **15**

Question 9 **17**

Question 10 **19**

Question 11 **21**

Question 12 **23**

Question 13 **25**

Question 14 **27**

Question 15 **29**

1. **(a)** Show that an appropriate integrating factor for

$$x^2 \frac{\mathrm{d}y}{\mathrm{d}x} + y = 1, \qquad x > 0$$

is $e^{-1/x}$ **[4]**

(b) Hence find the solution of the differential equation

$$x^2 \frac{\mathrm{d}y}{\mathrm{d}x} + y = 1, \qquad x > 0$$

for which $y = 3$ when $x = 1$ **[6]**

Solution

(a) First write the differential equation in the standard linear form by dividing through by x^2:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{x^2} y = \frac{1}{x^2}$$

This is of the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$$

with

$$P(x) = \frac{1}{x^2} = x^{-2}$$

For a linear differential equation, an integrating factor is

$$\mu(x) = e^{\int P(x)\,\mathrm{d}x}$$

So

$$\begin{aligned} \mu(x) &= e^{\int x^{-2}\,\mathrm{d}x} \\ &= e^{-x^{-1}} \\ &= e^{-1/x} \end{aligned}$$

Hence an appropriate integrating factor is $e^{-1/x}$.

(b) Using the integrating factor from part (a), multiply

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{x^2} y = \frac{1}{x^2}$$

through by $e^{-1/x}$:

$$e^{-1/x} \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{e^{-1/x}}{x^2} y = \frac{e^{-1/x}}{x^2}$$

Now apply the product rule. Since

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(e^{-1/x}\right) = e^{-1/x} \cdot \frac{1}{x^2}$$

we get

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}x}\left(ye^{-1/x}\right) &= e^{-1/x} \frac{\mathrm{d}y}{\mathrm{d}x} + y \frac{\mathrm{d}}{\mathrm{d}x}\left(e^{-1/x}\right) \\ &= e^{-1/x} \frac{\mathrm{d}y}{\mathrm{d}x} + \frac{e^{-1/x}}{x^2} y \end{aligned}$$

So the differential equation becomes

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(ye^{-1/x}\right) = \frac{e^{-1/x}}{x^2}$$

Integrating both sides with respect to x gives

$$ye^{-1/x} = \int \frac{e^{-1/x}}{x^2}\,\mathrm{d}x + C$$

But

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(e^{-1/x}\right) = \frac{e^{-1/x}}{x^2}$$

so

$$\int \frac{e^{-1/x}}{x^2}\,\mathrm{d}x = e^{-1/x}$$

Hence

$$ye^{-1/x} = e^{-1/x} + C$$

Multiplying through by $e^{1/x}$:

$$y = 1 + Ce^{1/x}$$

Now use the condition $y = 3$ when $x = 1$:

$$\begin{aligned} 3 &= 1 + Ce \\ 2 &= Ce \\ C &= \frac{2}{e} \end{aligned}$$

Substitute back:

$$\begin{aligned} y &= 1 + \frac{2}{e}e^{1/x} \\ &= 1 + 2e^{1/x-1} \end{aligned}$$

Therefore the required solution is $y = 1 + 2e^{1/x-1}$.

2. **(a)** Show that

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(\ln(\cosh x+1)\big)=\tanh\left(\frac{x}{2}\right)$$ **[3]**

(b) Hence solve the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x}+\tanh\left(\frac{x}{2}\right)y=\sinh x$$

for which $y=2$ when $x=0$. Give your answer in exact form **[7]**

Solution

(a) Using $\dfrac{\mathrm{d}}{\mathrm{d}x}\ln u=\dfrac{1}{u}\dfrac{\mathrm{d}u}{\mathrm{d}x}$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(\ln(\cosh x+1)\big)=\frac{1}{\cosh x+1}\cdot\frac{\mathrm{d}}{\mathrm{d}x}(\cosh x+1)=\frac{\sinh x}{\cosh x+1}$$

Now use the half-angle identities

$$\sinh x=2\sinh\left(\frac{x}{2}\right)\cosh\left(\frac{x}{2}\right),\qquad \cosh x+1=2\cosh^2\left(\frac{x}{2}\right)$$

So

$$\begin{aligned}\frac{\sinh x}{\cosh x+1}&=\frac{2\sinh\left(\frac{x}{2}\right)\cosh\left(\frac{x}{2}\right)}{2\cosh^2\left(\frac{x}{2}\right)}\\&=\frac{\sinh\left(\frac{x}{2}\right)}{\cosh\left(\frac{x}{2}\right)}\\&=\tanh\left(\frac{x}{2}\right)\end{aligned}$$

Hence

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(\ln(\cosh x+1)\big)=\tanh\left(\frac{x}{2}\right)$$

as required.

(b) The differential equation is linear:

$$\frac{\mathrm{d}y}{\mathrm{d}x}+\tanh\left(\frac{x}{2}\right)y=\sinh x$$

From part (a),

$$\frac{\mathrm{d}}{\mathrm{d}x}\big(\ln(\cosh x+1)\big)=\tanh\left(\frac{x}{2}\right)$$

so an integrating factor is

$$\mu(x)=\mathrm{e}^{\int\tanh\left(\frac{x}{2}\right)\mathrm{d}x}=\mathrm{e}^{\ln(\cosh x+1)}=\cosh x+1$$

Multiplying the differential equation throughout by $\cosh x+1$,

$$(\cosh x+1)\frac{\mathrm{d}y}{\mathrm{d}x}+(\cosh x+1)\tanh\left(\frac{x}{2}\right)y=(\cosh x+1)\sinh x$$

The left-hand side is

$$\frac{\mathrm{d}}{\mathrm{d}x}\big((\cosh x+1)y\big)$$

Hence the differential equation can be written

$$\frac{\mathrm{d}}{\mathrm{d}x}\big((\cosh x+1)y\big)=(\cosh x+1)\sinh x$$

We can integrate the right hand side by reversing the chain rule:

$$\int (\cosh x + 1)\sinh x \,\mathrm{d}x = \frac{1}{2}(\cosh x + 1)^2 + C$$

So the differential equation becomes

$$\begin{aligned}(\cosh x + 1)y &= \int (\cosh x + 1)\sinh x \,\mathrm{d}x \\ &= \frac{1}{2}(\cosh x + 1)^2 + C\end{aligned}$$

Therefore

$$y = \frac{1}{2}(\cosh x + 1) + \frac{C}{\cosh x + 1}$$

Use the condition $y = 2$ when $x = 0$. Since $\cosh 0 = 1$,

$$\begin{aligned}2 &= \frac{1}{2}(1+1) + \frac{C}{1+1} \\ 2 &= 1 + \frac{C}{2}\end{aligned}$$

Hence

$$C = 2$$

Substituting back,

$$y = \frac{\cosh x + 1}{2} + \frac{2}{\cosh x + 1}$$

Therefore the exact solution is

$$\boxed{y = \frac{\cosh x + 1}{2} + \frac{2}{\cosh x + 1}}$$

Alternatively, one can integrate using the double angle formula for $\sinh 2x$:

$$\int (\cosh x + 1)\sinh x \,\mathrm{d}x = \int \cosh x \sinh x + \cosh x \,\mathrm{d}x = \int \frac{1}{2}\sinh 2x + \cosh x \,\mathrm{d}x$$

to obtain an equivalent form of the final answer:

$$\boxed{y = \frac{\frac{1}{4}\cosh 2x + \cosh x + \frac{11}{4}}{\cosh x + 1}}$$

3. **(a)** For $x > -1$, determine the general solution of the differential equation

$$(1+x)\frac{\mathrm{d}y}{\mathrm{d}x} + y = \ln(1+x)$$

giving your answer in the form $y = f(x)$ **[4]**

(b) Given that $y = -1$ when $x = 0$, determine the positive value of x for which $y = 0$ **[2]**

Solution

(a) For $x > -1$, divide the differential equation by $1 + x$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{1}{1+x}y = \frac{\ln(1+x)}{1+x}$$

This is a linear differential equation of the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$$

with

$$P(x) = \frac{1}{1+x}$$

The integrating factor is

$$\begin{aligned}\mu(x) &= e^{\int \frac{1}{1+x}\,\mathrm{d}x} \\ &= e^{\ln(1+x)}\end{aligned}$$

Since $x > -1$, we have $1 + x > 0$, so

$$\mu(x) = 1 + x$$

Multiplying the differential equation by $1 + x$ gives

$$(1+x)\frac{\mathrm{d}y}{\mathrm{d}x} + y = \ln(1+x)$$

The left-hand side is the derivative of $(1 + x)y$:

$$\frac{\mathrm{d}}{\mathrm{d}x}\big((1+x)y\big) = \ln(1+x)$$

Integrating both sides with respect to x:

$$(1+x)y = \int \ln(1+x)\,\mathrm{d}x + C$$

Now evaluate the integral by letting $u = 1 + x$, so $\mathrm{d}u = \mathrm{d}x$:

$$\begin{aligned}\int \ln(1+x)\,\mathrm{d}x &= \int \ln u\,\mathrm{d}u \\ &= u\ln u - u + C \\ &= (1+x)\ln(1+x) - (1+x) + C\end{aligned}$$

So

$$\begin{aligned}(1+x)y &= (1+x)\ln(1+x) - (1+x) + C \\ &= (1+x)\ln(1+x) - x - 1 + C\end{aligned}$$

Absorbing the constant -1 into C, we write

$$(1+x)y = (1+x)\ln(1+x) - x + C$$

Hence

$$\begin{aligned} y &= \frac{(1+x)\ln(1+x) - x + C}{1+x} \\ &= \ln(1+x) - \frac{x}{1+x} + \frac{C}{1+x} \\ &= \ln(1+x) - 1 + \frac{1}{1+x} + \frac{C}{1+x} \end{aligned}$$

Absorbing the extra constant into C again gives

$$y = \ln(1+x) - 1 + \frac{C}{1+x}$$

Therefore, the general solution is

$$y = \ln(1+x) - 1 + \frac{C}{1+x}$$

(b) Use the condition $y = -1$ when $x = 0$.

Substituting into the general solution:

$$\begin{aligned} -1 &= \ln(1+0) - 1 + \frac{C}{1+0} \\ -1 &= 0 - 1 + C \\ -1 &= -1 + C \end{aligned}$$

So

$$C = 0$$

Hence

$$y = \ln(1+x) - 1$$

Now set $y = 0$:

$$\begin{aligned} 0 &= \ln(1+x) - 1 \\ \ln(1+x) &= 1 \end{aligned}$$

Exponentiating both sides:

$$\begin{aligned} 1 + x &= e \\ x &= e - 1 \end{aligned}$$

Since the question asks for the positive value,

$$x = e - 1 \approx 1.72$$

Therefore, the required value is $x = e - 1 \approx 1.72$.

4. **(a)** Show that, for $0 < x < \frac{\pi}{2}$, an appropriate integrating factor for

$$\cos x \frac{\mathrm{d}y}{\mathrm{d}x} + y \sin x = x$$

is $\sec x$ **[4]**

(b) Hence find the solution of

$$\cos x \frac{\mathrm{d}y}{\mathrm{d}x} + y \sin x = x$$

for $0 < x < \frac{\pi}{2}$, given that $y = 1$ when $x = \frac{\pi}{4}$ **[6]**

Solution

(a) First divide the differential equation by $\cos x$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + y \tan x = x \sec x$$

This is now in the linear form

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$$

with

$$P(x) = \tan x$$

So an integrating factor is

$$\begin{aligned}
\mu(x) &= e^{\int P(x)\,\mathrm{d}x} \\
&= e^{\int \tan x\,\mathrm{d}x} \\
&= e^{\int \frac{\sin x}{\cos x}\,\mathrm{d}x} \\
&= e^{-\ln|\cos x|} \\
&= e^{\ln|\sec x|}
\end{aligned}$$

Since $0 < x < \frac{\pi}{2}$, we have $\cos x > 0$, so $|\sec x| = \sec x$. Hence

$$\mu(x) = e^{\ln(\sec x)} = \sec x$$

Therefore an appropriate integrating factor is $\sec x$.

(b) Using the result from part (a), the equation is

$$\frac{\mathrm{d}y}{\mathrm{d}x} + y \tan x = x \sec x$$

Multiply through by the integrating factor $\sec x$:

$$\sec x \frac{\mathrm{d}y}{\mathrm{d}x} + y \sec x \tan x = x \sec^2 x$$

The left-hand side is the derivative of $y \sec x$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}(y \sec x) = x \sec^2 x$$

Integrating both sides gives

$$y \sec x = \int x \sec^2 x\,\mathrm{d}x + C$$

Integrate by parts with

$$u = x, \qquad \mathrm{d}v = \sec^2 x\,\mathrm{d}x$$

so that

$$\mathrm{d}u = \mathrm{d}x, \qquad v = \tan x$$

Then

$$\begin{aligned}\int x\sec^2 x\,\mathrm{d}x &= x\tan x - \int \tan x\,\mathrm{d}x \\ &= x\tan x - (-\ln(\cos x)) \\ &= x\tan x + \ln(\cos x)\end{aligned}$$

Therefore

$$y\sec x = x\tan x + \ln(\cos x) + C$$

Now multiply through by $\cos x$:

$$\begin{aligned}y &= \cos x\,(x\tan x + \ln(\cos x) + C) \\ &= x\sin x + \cos x\ln(\cos x) + C\cos x\end{aligned}$$

Use the condition $y = 1$ when $x = \frac{\pi}{4}$. Substituting into

$$y\sec x = x\tan x + \ln(\cos x) + C$$

gives

$$\begin{aligned}1\cdot\sec\frac{\pi}{4} &= \frac{\pi}{4}\tan\frac{\pi}{4} + \ln\left(\cos\frac{\pi}{4}\right) + C \\ \sqrt{2} &= \frac{\pi}{4} + \ln\left(\frac{\sqrt{2}}{2}\right) + C \\ \sqrt{2} &= \frac{\pi}{4} - \frac{1}{2}\ln 2 + C\end{aligned}$$

So

$$C = \sqrt{2} - \frac{\pi}{4} + \frac{1}{2}\ln 2$$

Substitute this into the expression for y:

$$y = x\sin x + \cos x\left[\ln(\cos x) + \sqrt{2} - \frac{\pi}{4} + \frac{1}{2}\ln 2\right]$$

Hence the required solution is

$$y = x\sin x + \cos x\left[\ln(\cos x) + \sqrt{2} - \frac{\pi}{4} + \frac{1}{2}\ln 2\right]$$

5. **(a)** Show that, for $x > 0$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\coth\frac{x}{2}\right)\right) = -\operatorname{cosech} x$$ **[3]**

(b) Hence solve the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} - \operatorname{cosech} x\, y = \tanh\frac{x}{2}, \qquad x > 0$$

for which $y = 1$ when $x = \ln 3$. Give your answer in exact form **[7]**

Solution

(a) Let

$$u = \frac{x}{2}$$

Then, by the chain rule,

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\coth\frac{x}{2}\right)\right) &= \frac{\mathrm{d}}{\mathrm{d}x}\big(\ln(\coth u)\big)\\ &= \frac{1}{\coth u}\cdot\frac{\mathrm{d}}{\mathrm{d}x}(\coth u)\\ &= \frac{1}{\coth u}\cdot\left(-\operatorname{cosech}^2 u\right)\cdot\frac{1}{2}\end{aligned}$$

Now simplify:

$$\begin{aligned}\frac{1}{\coth u}\cdot\left(-\operatorname{cosech}^2 u\right)\cdot\frac{1}{2} &= -\frac{1}{2}\cdot\frac{\operatorname{cosech}^2 u}{\coth u}\\ &= -\frac{1}{2}\cdot\frac{1/\sinh^2 u}{\cosh u/\sinh u}\\ &= -\frac{1}{2\sinh u\cosh u}\end{aligned}$$

Using $\sinh(2u) = 2\sinh u\cosh u$,

$$\begin{aligned}-\frac{1}{2\sinh u\cosh u} &= -\frac{1}{\sinh(2u)}\\ &= -\frac{1}{\sinh x}\\ &= -\operatorname{cosech} x\end{aligned}$$

Therefore,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\coth\frac{x}{2}\right)\right) = -\operatorname{cosech} x$$

as required.

(b) The differential equation is

$$\frac{\mathrm{d}y}{\mathrm{d}x} - \operatorname{cosech} x\, y = \tanh\frac{x}{2}$$

This is linear, with integrating factor

$$\begin{aligned}\mu(x) &= e^{\int -\operatorname{cosech} x\,\mathrm{d}x}\\ &= e^{\ln\left(\coth\frac{x}{2}\right)}\\ &= \coth\frac{x}{2}\end{aligned}$$

using part (a).

Next, from part (a),

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\coth\frac{x}{2}\right)\right) = -\operatorname{cosech} x$$

so

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left(\coth\frac{x}{2}\right) &= \coth\frac{x}{2}\cdot\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\coth\frac{x}{2}\right)\right)\\ &= -\operatorname{cosech} x\,\coth\frac{x}{2}\end{aligned}$$

Multiply the differential equation by $\coth\frac{x}{2}$:

$$\coth\frac{x}{2}\,\frac{\mathrm{d}y}{\mathrm{d}x} - \operatorname{cosech} x\,\coth\frac{x}{2}\,y = \coth\frac{x}{2}\tanh\frac{x}{2}$$

The left-hand side is

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(y\coth\frac{x}{2}\right)$$

and the right-hand side is 1, since $\coth\frac{x}{2}\tanh\frac{x}{2} = 1$. Hence

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(y\coth\frac{x}{2}\right) = 1$$

Integrating,

$$y\coth\frac{x}{2} = \int 1\,\mathrm{d}x = x + C$$

So

$$y = (x+C)\tanh\frac{x}{2}$$

Now use $y = 1$ when $x = \ln 3$:

$$1 = (\ln 3 + C)\tanh\frac{\ln 3}{2}$$

Evaluate $\tanh\frac{\ln 3}{2}$:

$$\begin{aligned}\tanh\frac{\ln 3}{2} &= \frac{e^{\ln 3/2} - e^{-\ln 3/2}}{e^{\ln 3/2} + e^{-\ln 3/2}}\\ &= \frac{e^{\ln 3} - 1}{e^{\ln 3} + 1}\\ &= \frac{3-1}{3+1}\\ &= \frac{1}{2}\end{aligned}$$

Therefore,

$$\begin{aligned}1 &= (\ln 3 + C)\cdot\frac{1}{2}\\ 2 &= \ln 3 + C\\ C &= 2 - \ln 3\end{aligned}$$

Substituting back gives

$$y = (x + 2 - \ln 3)\tanh\frac{x}{2}$$

Therefore, the required exact solution is $y = (x + 2 - \ln 3)\tanh\frac{x}{2}$.

6. **(a)** Show that an appropriate integrating factor for

$$x(x-1)\frac{\mathrm{d}y}{\mathrm{d}x} + (x+1)y = x(x-1)^2$$

is $\dfrac{(x-1)^2}{x}$ **[4]**

(b) Hence find the solution of the differential equation

$$x(x-1)\frac{\mathrm{d}y}{\mathrm{d}x} + (x+1)y = x(x-1)^2$$

for $x > 1$, given that $y = 2$ when $x = 2$ **[6]**

Solution

(a) First write the differential equation in linear form by dividing through by $x(x-1)$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{x+1}{x(x-1)}y = x-1$$

For a linear equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x),$$

an integrating factor is

$$\mu(x) = e^{\int P(x)\,\mathrm{d}x}$$

Here

$$P(x) = \frac{x+1}{x(x-1)}$$

Decompose into partial fractions:

$$\begin{aligned}\frac{x+1}{x(x-1)} &= \frac{A}{x} + \frac{B}{x-1}\\ x+1 &= A(x-1) + Bx\\ x+1 &= (A+B)x - A\end{aligned}$$

Equating coefficients gives

$$A + B = 1, \qquad -A = 1$$

so

$$A = -1, \qquad B = 2$$

Hence

$$\frac{x+1}{x(x-1)} = -\frac{1}{x} + \frac{2}{x-1}$$

Therefore

$$\begin{aligned}\mu(x) &= e^{\int\left(-\frac{1}{x}+\frac{2}{x-1}\right)\mathrm{d}x}\\ &= e^{-\ln x + 2\ln(x-1)}\\ &= \frac{(x-1)^2}{x}\end{aligned}$$

Thus an appropriate integrating factor is

$$\mu(x) = \frac{(x-1)^2}{x}$$

(b) Using the integrating factor from part (a),

$$\mu(x) = \frac{(x-1)^2}{x}$$

Multiply the linear form

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{x+1}{x(x-1)}y = x - 1$$

through by $\mu(x)$:

$$\frac{(x-1)^2}{x}\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{(x-1)^2}{x} \cdot \frac{x+1}{x(x-1)}\,y = \frac{(x-1)^2}{x}(x-1)$$

Since μ is an integrating factor, the left-hand side becomes

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{(x-1)^2}{x}y\right)$$

So

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{(x-1)^2}{x}y\right) = \frac{(x-1)^3}{x}$$

Integrating both sides:

$$\frac{(x-1)^2}{x}y = \int \frac{(x-1)^3}{x}\,\mathrm{d}x$$

Expand the integrand:

$$\begin{aligned}\frac{(x-1)^3}{x} &= \frac{x^3 - 3x^2 + 3x - 1}{x} \\ &= x^2 - 3x + 3 - \frac{1}{x}\end{aligned}$$

Hence

$$\begin{aligned}\frac{(x-1)^2}{x}y &= \int \left(x^2 - 3x + 3 - \frac{1}{x}\right)\,\mathrm{d}x \\ &= \frac{x^3}{3} - \frac{3x^2}{2} + 3x - \ln x + C\end{aligned}$$

So

$$y = \frac{x}{(x-1)^2}\left(\frac{x^3}{3} - \frac{3x^2}{2} + 3x - \ln x + C\right)$$

Now use the condition $y = 2$ when $x = 2$:

$$\begin{aligned}2 &= \frac{2}{(2-1)^2}\left(\frac{2^3}{3} - \frac{3(2^2)}{2} + 3(2) - \ln 2 + C\right) \\ 2 &= 2\left(\frac{8}{3} - 6 + 6 - \ln 2 + C\right) \\ 2 &= 2\left(\frac{8}{3} - \ln 2 + C\right)\end{aligned}$$

Divide by 2:

$$1 = \frac{8}{3} - \ln 2 + C$$

So

$$C = 1 - \frac{8}{3} + \ln 2 = \ln 2 - \frac{5}{3}$$

Substituting back:

$$y = \frac{x}{(x-1)^2}\left(\frac{x^3}{3} - \frac{3x^2}{2} + 3x - \ln x + \ln 2 - \frac{5}{3}\right)$$

Therefore the required solution is

$$y = \frac{x}{(x-1)^2}\left(\frac{x^3}{3} - \frac{3x^2}{2} + 3x - \ln x + \ln 2 - \frac{5}{3}\right)$$

7. Find the general solution of the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \cot x\, y = \frac{\cos x}{\sin^2 x\left(1+\ln^2(\sin x)\right)}$$

where $0 < x < \pi$ **[7]**

Solution

This is a first-order linear differential equation:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \cot x\, y = \frac{\cos x}{\sin^2 x\left(1+\ln^2(\sin x)\right)}$$

Its integrating factor is

$$\begin{aligned}\mu(x) &= e^{\int \cot x\,\mathrm{d}x}\\ &= e^{\int \frac{\cos x}{\sin x}\,\mathrm{d}x}\\ &= e^{\ln(\sin x)}\end{aligned}$$

Since $0 < x < \pi$, we have $\sin x > 0$, so

$$\mu(x) = \sin x$$

Multiply the whole differential equation by $\sin x$:

$$\sin x\frac{\mathrm{d}y}{\mathrm{d}x} + \sin x\cot x\, y = \sin x\cdot\frac{\cos x}{\sin^2 x\left(1+\ln^2(\sin x)\right)}$$

$$\sin x\frac{\mathrm{d}y}{\mathrm{d}x} + \cos x\, y = \frac{\cos x}{\sin x\left(1+\ln^2(\sin x)\right)}$$

Now

$$\frac{\mathrm{d}}{\mathrm{d}x}(y\sin x) = \sin x\frac{\mathrm{d}y}{\mathrm{d}x} + \cos x\, y$$

So the equation becomes

$$\frac{\mathrm{d}}{\mathrm{d}x}(y\sin x) = \frac{\cos x}{\sin x\left(1+\ln^2(\sin x)\right)}$$

Integrating both sides gives

$$\int \frac{\mathrm{d}}{\mathrm{d}x}(y\sin x)\ \mathrm{d}x = \int \frac{\cos x}{\sin x\left(1+\ln^2(\sin x)\right)}\ \mathrm{d}x$$

For the right-hand side, let

$$u = \ln(\sin x)$$

Then

$$\begin{aligned}\frac{\mathrm{d}u}{\mathrm{d}x} &= \frac{1}{\sin x}\cdot\cos x\\ &= \frac{\cos x}{\sin x}\end{aligned}$$

Hence

$$\mathrm{d}u = \frac{\cos x}{\sin x}\,\mathrm{d}x$$

So

$$\begin{aligned}\int \frac{\cos x}{\sin x\left(1+\ln^2(\sin x)\right)}\ \mathrm{d}x &= \int \frac{1}{1+u^2}\,\mathrm{d}u\\ &= \arctan(u) + C\\ &= \arctan(\ln(\sin x)) + C\end{aligned}$$

Therefore

$$y\sin x = \arctan(\ln(\sin x)) + C$$

Finally, divide by $\sin x$:

$$y = \frac{\arctan(\ln(\sin x)) + C}{\sin x}$$

The general solution is $y = \dfrac{\arctan(\ln(\sin x)) + C}{\sin x}$.

8. **(a)** Solve the differential equation

$$(1+x)\frac{\mathrm{d}y}{\mathrm{d}x} - 2y = (1+x)^4, \qquad x > -1$$

giving your answer in the form $y = f(x)$ **[3]**

(b) The solution curve passes through the point $(0, -1)$. Find the exact positive value of x for which $y = 0$ **[3]**

Solution

(a) Rewrite the differential equation in linear form by dividing through by $1+x$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{2}{1+x}y = (1+x)^3$$

Since this is a linear equation, the integrating factor is

$$\begin{aligned}\mu(x) &= e^{\int -\frac{2}{1+x}\,\mathrm{d}x} \\ &= e^{-2\ln(1+x)} \\ &= (1+x)^{-2}\end{aligned}$$

Multiplying the differential equation by $(1+x)^{-2}$ gives

$$(1+x)^{-2}\frac{\mathrm{d}y}{\mathrm{d}x} - 2(1+x)^{-3}y = (1+x)$$

The left-hand side is the derivative of $y(1+x)^{-2}$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(y(1+x)^{-2}\right) = 1+x$$

Integrating both sides:

$$\begin{aligned}y(1+x)^{-2} &= \int (1+x)\,\mathrm{d}x \\ &= x + \frac{x^2}{2} + C\end{aligned}$$

Hence

$$y = (1+x)^2\left(\frac{x^2}{2} + x + C\right)$$

Now use the point $(0, -1)$:

$$\begin{aligned}-1 &= (1+0)^2\left(\frac{0^2}{2} + 0 + C\right) \\ -1 &= C\end{aligned}$$

Therefore

$$y = (1+x)^2\left(\frac{x^2}{2} + x - 1\right)$$

So the required function is $y = (1+x)^2\left(\frac{x^2}{2} + x - 1\right)$.

(b) To find when $y = 0$, set the expression from part (a) equal to zero:

$$(1+x)^2\left(\frac{x^2}{2} + x - 1\right) = 0$$

Since $x > -1$, we have $1 + x > 0$, so $(1 + x)^2 \neq 0$. Therefore

$$\frac{x^2}{2} + x - 1 = 0$$

Multiply by 2:

$$x^2 + 2x - 2 = 0$$

Solve using the quadratic formula:

$$\begin{aligned} x &= \frac{-2 \pm \sqrt{2^2 - 4(1)(-2)}}{2} \\ &= \frac{-2 \pm \sqrt{4 + 8}}{2} \\ &= \frac{-2 \pm \sqrt{12}}{2} \\ &= \frac{-2 \pm 2\sqrt{3}}{2} \\ &= -1 \pm \sqrt{3} \end{aligned}$$

The positive value is

$$x = \sqrt{3} - 1$$

So the exact positive value of x is $x = \sqrt{3} - 1$.

9. Find the general solution of the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 2y\tan x = \sec^3 x$$

where $-\frac{\pi}{2} < x < \frac{\pi}{2}$ **[7]**

Solution

Write the differential equation in the linear form

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$$

so here

$$\frac{\mathrm{d}y}{\mathrm{d}x} - 2y\tan x = \sec^3 x, \qquad P(x) = -2\tan x, \qquad Q(x) = \sec^3 x$$

For a linear equation, the integrating factor is

$$\mu(x) = e^{\int P(x)\ \mathrm{d}x}$$

So

$$\begin{aligned}\mu(x) &= e^{\int -2\tan x\ \mathrm{d}x}\\ &= e^{-2\int \frac{\sin x}{\cos x}\ \mathrm{d}x}\\ &= e^{-2(-\ln(\cos x))}\\ &= e^{2\ln(\cos x)}\\ &= \cos^2 x\end{aligned}$$

Since $-\frac{\pi}{2} < x < \frac{\pi}{2}$, we have $\cos x > 0$, so this is valid without absolute values.
Now multiply the whole differential equation by $\cos^2 x$:

$$\begin{aligned}\cos^2 x\frac{\mathrm{d}y}{\mathrm{d}x} - 2y\cos^2 x\tan x &= \cos^2 x\sec^3 x\\ \cos^2 x\frac{\mathrm{d}y}{\mathrm{d}x} - 2y\cos x\sin x &= \sec x\end{aligned}$$

Next, observe that

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}(y\cos^2 x) &= \cos^2 x\frac{\mathrm{d}y}{\mathrm{d}x} + y\frac{\mathrm{d}}{\mathrm{d}x}(\cos^2 x)\\ &= \cos^2 x\frac{\mathrm{d}y}{\mathrm{d}x} + y(2\cos x)(-\sin x)\\ &= \cos^2 x\frac{\mathrm{d}y}{\mathrm{d}x} - 2y\cos x\sin x\end{aligned}$$

So the left-hand side is an exact derivative, and the equation becomes

$$\frac{\mathrm{d}}{\mathrm{d}x}(y\cos^2 x) = \sec x$$

Integrating both sides with respect to x gives

$$y\cos^2 x = \int \sec x\ \mathrm{d}x$$

To integrate $\sec x$, use the standard trick:

$$\begin{aligned}\int \sec x\ \mathrm{d}x &= \int \sec x\cdot\frac{\sec x + \tan x}{\sec x + \tan x}\ \mathrm{d}x\\ &= \int \frac{\sec^2 x + \sec x\tan x}{\sec x + \tan x}\ \mathrm{d}x\end{aligned}$$

Let

$$u = \sec x + \tan x$$

Then

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \sec x \tan x + \sec^2 x$$

so

$$\int \sec x \ \mathrm{d}x = \int \frac{1}{u} \ \mathrm{d}u = \ln|u| + C = \ln|\sec x + \tan x| + C$$

On the interval $-\frac{\pi}{2} < x < \frac{\pi}{2}$, we have

$$\sec x + \tan x = \frac{1 + \sin x}{\cos x} > 0$$

because $1 + \sin x > 0$ and $\cos x > 0$. Hence

$$\ln|\sec x + \tan x| = \ln(\sec x + \tan x)$$

Therefore

$$y \cos^2 x = \ln(\sec x + \tan x) + C$$

Finally, divide through by $\cos^2 x$:

$$y = \sec^2 x\big(\ln(\sec x + \tan x) + C\big)$$

Hence the general solution is

$$y = \sec^2 x\big(\ln(\sec x + \tan x) + C\big)$$

10. **(a)** Show that

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\frac{\cosh x+1}{\sinh x}\right)\right) = -\operatorname{cosech} x$$ **[3]**

(b) Hence solve, for $x > 0$,

$$\sinh x\,\frac{\mathrm{d}y}{\mathrm{d}x} - y = 1 - \cosh x$$

given that $y = 2$ when $x = \ln 3$. Give your answer in exact form **[7]**

Solution

(a) Write

$$\ln\left(\frac{\cosh x+1}{\sinh x}\right) = \ln(\cosh x+1) - \ln(\sinh x)$$

Differentiate term by term:

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\frac{\cosh x+1}{\sinh x}\right)\right) &= \frac{\sinh x}{\cosh x+1} - \frac{\cosh x}{\sinh x}\\ &= \frac{\sinh^2 x - \cosh x(\cosh x+1)}{\sinh x(\cosh x+1)}\end{aligned}$$

Now simplify the numerator using $\cosh^2 x - \sinh^2 x = 1$:

$$\begin{aligned}\sinh^2 x - \cosh x(\cosh x+1) &= \sinh^2 x - \cosh^2 x - \cosh x\\ &= -(\cosh^2 x - \sinh^2 x) - \cosh x\\ &= -1 - \cosh x\\ &= -(\cosh x+1)\end{aligned}$$

So

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\frac{\cosh x+1}{\sinh x}\right)\right) &= \frac{-(\cosh x+1)}{\sinh x(\cosh x+1)}\\ &= -\frac{1}{\sinh x}\\ &= -\operatorname{cosech} x\end{aligned}$$

Hence,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\frac{\cosh x+1}{\sinh x}\right)\right) = -\operatorname{cosech} x$$

(b) Start with

$$\sinh x\,\frac{\mathrm{d}y}{\mathrm{d}x} - y = 1 - \cosh x$$

Since $x > 0$, we have $\sinh x \neq 0$, so divide through by $\sinh x$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} - \operatorname{cosech} x\, y = \frac{1-\cosh x}{\sinh x}$$

This is a linear differential equation of the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$$

with

$$P(x) = -\operatorname{cosech} x$$

From part (a),

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\ln\left(\frac{\cosh x+1}{\sinh x}\right)\right) = -\operatorname{cosech} x$$

so an integrating factor is

$$\mu(x) = \exp\left(\int -\operatorname{cosech} x \, \mathrm{d}x\right) = \frac{\cosh x + 1}{\sinh x}$$

Multiply the differential equation throughout by $\mu(x)$:

$$\frac{\cosh x + 1}{\sinh x}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{\cosh x + 1}{\sinh^2 x}y = \frac{(\cosh x + 1)(1 - \cosh x)}{\sinh^2 x}$$

The left-hand side is

$$\frac{\mathrm{d}}{\mathrm{d}x}(\mu y) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\cosh x + 1}{\sinh x}y\right)$$

Simplify the right-hand side:

$$\begin{aligned}\frac{(\cosh x + 1)(1 - \cosh x)}{\sinh^2 x} &= \frac{1 - \cosh^2 x}{\sinh^2 x} \\ &= \frac{-\sinh^2 x}{\sinh^2 x} \\ &= -1\end{aligned}$$

So

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\cosh x + 1}{\sinh x}y\right) = -1$$

Integrate with respect to x:

$$\frac{\cosh x + 1}{\sinh x}y = -x + C$$

Hence

$$y = \frac{(C - x)\sinh x}{\cosh x + 1}$$

Now use the condition $y = 2$ when $x = \ln 3$.
First find $\sinh(\ln 3)$ and $\cosh(\ln 3)$:

$$\begin{aligned}\sinh(\ln 3) &= \frac{e^{\ln 3} - e^{-\ln 3}}{2} = \frac{3 - \frac{1}{3}}{2} = \frac{4}{3} \\ \cosh(\ln 3) &= \frac{e^{\ln 3} + e^{-\ln 3}}{2} = \frac{3 + \frac{1}{3}}{2} = \frac{5}{3}\end{aligned}$$

Substitute into the solution:

$$\begin{aligned}2 &= \frac{(C - \ln 3)\sinh(\ln 3)}{\cosh(\ln 3) + 1} \\ &= \frac{(C - \ln 3)\cdot\frac{4}{3}}{\frac{5}{3} + 1} \\ &= \frac{(C - \ln 3)\cdot\frac{4}{3}}{\frac{8}{3}} \\ &= \frac{C - \ln 3}{2}\end{aligned}$$

Therefore

$$\begin{aligned}2 &= \frac{C - \ln 3}{2} \\ 4 &= C - \ln 3 \\ C &= 4 + \ln 3\end{aligned}$$

Substituting back,

$$y = \frac{(4 + \ln 3 - x)\sinh x}{\cosh x + 1}$$

So the exact solution is

$$y = \frac{(4 + \ln 3 - x)\sinh x}{\cosh x + 1}$$

11. **(a)** Find the general solution of the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2x}{x^2+1}y = 6x$$

giving your answer in the form $y = f(x)$ **[3]**

(b) Given that $y = \frac{7}{2}$ when $x = 0$, find the smallest positive value of x for which $y = 4$ **[3]**

Solution

(a) The differential equation is linear:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + \frac{2x}{x^2+1}y = 6x$$

Its integrating factor is

$$\begin{aligned}\mu(x) &= e^{\int \frac{2x}{x^2+1}\,\mathrm{d}x} \\ &= e^{\ln(x^2+1)} \\ &= x^2+1\end{aligned}$$

Multiplying the differential equation by x^2+1 gives

$$(x^2+1)\frac{\mathrm{d}y}{\mathrm{d}x} + 2xy = 6x(x^2+1)$$

The left-hand side is the derivative of $(x^2+1)y$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\big((x^2+1)y\big) = 6x(x^2+1)$$

Integrating both sides:

$$\begin{aligned}(x^2+1)y &= \int 6x(x^2+1)\,\mathrm{d}x \\ &= \int (6x^3+6x)\,\mathrm{d}x \\ &= \frac{3}{2}x^4 + 3x^2 + C\end{aligned}$$

Hence

$$y = \frac{\frac{3}{2}x^4 + 3x^2 + C}{x^2+1}$$

Therefore, the general solution is

$$y = \frac{\frac{3}{2}x^4 + 3x^2 + C}{x^2+1}$$

(b) Using $y = \frac{7}{2}$ when $x = 0$:

$$\begin{aligned}\frac{7}{2} &= \frac{\frac{3}{2}(0)^4 + 3(0)^2 + C}{(0)^2+1} \\ \frac{7}{2} &= C\end{aligned}$$

So the particular solution is

$$y = \frac{\frac{3}{2}x^4 + 3x^2 + \frac{7}{2}}{x^2+1}$$

Now set $y = 4$:

$$4 = \frac{\frac{3}{2}x^4 + 3x^2 + \frac{7}{2}}{x^2+1}$$

Multiply through by $x^2 + 1$:

$$4(x^2 + 1) = \frac{3}{2}x^4 + 3x^2 + \frac{7}{2}$$

Multiply by 2:

$$8x^2 + 8 = 3x^4 + 6x^2 + 7$$

Rearranging:

$$0 = 3x^4 - 2x^2 - 1$$

Let $u = x^2$. Then

$$\begin{aligned} 3u^2 - 2u - 1 &= 0 \\ (3u + 1)(u - 1) &= 0 \end{aligned}$$

So

$$u = 1 \quad \text{or} \quad u = -\frac{1}{3}$$

Since $u = x^2 \geq 0$, we take $x^2 = 1$, giving

$$x = \pm 1$$

The smallest positive value is

$$x = 1$$

12. For $x > 0$, solve the differential equation

$$\sinh x \frac{\mathrm{d}y}{\mathrm{d}x} - 2y \cosh x = \sinh^2 x \cosh x$$

given that $y = 5$ when $x = \ln(1 + \sqrt{2})$.
Give your answer in exact form. **[7]**

Solution

Since $x > 0$, we have $\sinh x > 0$, so in particular $\sinh x \neq 0$. We may therefore divide the differential equation by $\sinh x$ to put it into linear form:

$$\begin{aligned}
\sinh x \frac{\mathrm{d}y}{\mathrm{d}x} - 2y \cosh x &= \sinh^2 x \cosh x \\
\frac{\mathrm{d}y}{\mathrm{d}x} - 2\frac{\cosh x}{\sinh x} y &= \sinh x \cosh x \\
\frac{\mathrm{d}y}{\mathrm{d}x} - 2 \coth x \, y &= \sinh x \cosh x
\end{aligned}$$

This is a first-order linear equation of the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$$

with

$$P(x) = -2 \coth x$$

So the integrating factor is

$$\begin{aligned}
\mu(x) &= e^{\int -2 \coth x \, \mathrm{d}x} \\
&= e^{-2 \int \coth x \, \mathrm{d}x} \\
&= e^{-2 \ln(\sinh x)} \\
&= \frac{1}{\sinh^2 x}
\end{aligned}$$

Multiplying the differential equation by this integrating factor gives

$$\begin{aligned}
\frac{1}{\sinh^2 x} \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{2 \coth x}{\sinh^2 x} y &= \frac{\sinh x \cosh x}{\sinh^2 x} \\
\frac{\mathrm{d}}{\mathrm{d}x} \left(\frac{y}{\sinh^2 x} \right) &= \coth x
\end{aligned}$$

Now integrate both sides with respect to x:

$$\begin{aligned}
\frac{y}{\sinh^2 x} &= \int \coth x \, \mathrm{d}x \\
&= \ln(\sinh x) + C
\end{aligned}$$

Hence

$$y = \sinh^2 x \big(\ln(\sinh x) + C \big)$$

Now use the condition $y = 5$ when $x = \ln(1 + \sqrt{2})$.
We need $\sinh\big(\ln(1 + \sqrt{2})\big)$. Using

$$\sinh t = \frac{e^t - e^{-t}}{2}$$

with $t = \ln(1 + \sqrt{2})$, we get

$$\begin{aligned}
\sinh\big(\ln(1 + \sqrt{2})\big) &= \frac{(1 + \sqrt{2}) - \dfrac{1}{1 + \sqrt{2}}}{2} \\
&= \frac{(1 + \sqrt{2}) - (\sqrt{2} - 1)}{2} \\
&= \frac{2}{2} \\
&= 1
\end{aligned}$$

So substituting into the solution:

$$\begin{aligned} 5 &= 1^2\big(\ln 1 + C\big) \\ 5 &= 0 + C \\ C &= 5 \end{aligned}$$

Therefore the required exact solution is

$$y = \sinh^2 x\big(\ln(\sinh x) + 5\big)$$

So the answer is $y = \sinh^2 x\big(\ln(\sinh x) + 5\big)$.

13. For $x > 0$, a function y satisfies

$$\sinh x \frac{\mathrm{d}y}{\mathrm{d}x} - y \cosh x = \sinh x \cosh^3 x$$

(a) By first writing the equation in the form $\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$, find an integrating factor **[2]**

(b) Hence solve the differential equation, given that $y = 2$ when $x = \ln(1 + \sqrt{2})$ **[5]**

Give your answer in exact form.

Solution

(a) First divide the differential equation by $\sinh x$:

$$\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{\cosh x}{\sinh x} y = \cosh^3 x$$

So it is in the form $\dfrac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$ with

$$P(x) = -\coth x$$

The integrating factor is

$$\begin{aligned}\mu(x) &= e^{\int P(x)\ \mathrm{d}x} \\ &= e^{\int -\coth x\ \mathrm{d}x} \\ &= e^{-\ln(\sinh x)} \\ &= \frac{1}{\sinh x}\end{aligned}$$

since $x > 0$, so $\sinh x > 0$.

Final answer: the integrating factor is $\dfrac{1}{\sinh x}$.

(b) Multiply the equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} - \coth x\, y = \cosh^3 x$$

by the integrating factor $\dfrac{1}{\sinh x}$:

$$\frac{1}{\sinh x}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{\cosh x}{\sinh^2 x} y = \frac{\cosh^3 x}{\sinh x}$$

Now

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{\sinh x}\right) = \frac{1}{\sinh x}\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{\cosh x}{\sinh^2 x} y$$

so the equation becomes

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{\sinh x}\right) = \frac{\cosh^3 x}{\sinh x}$$

Integrating both sides gives

$$\frac{y}{\sinh x} = \int \frac{\cosh^3 x}{\sinh x}\ \mathrm{d}x + C$$

Rewrite the integrand using $\cosh^2 x = 1 + \sinh^2 x$:

$$\begin{aligned}\frac{\cosh^3 x}{\sinh x} &= \frac{\cosh x \cosh^2 x}{\sinh x} \\ &= \frac{\cosh x(1 + \sinh^2 x)}{\sinh x}\end{aligned}$$

Let

$$u = \sinh x \quad \Rightarrow \quad \mathrm{d}u = \cosh x \, \mathrm{d}x$$

Then

$$\begin{aligned}
\int \frac{\cosh^3 x}{\sinh x} \, \mathrm{d}x &= \int \frac{\cosh x (1 + \sinh^2 x)}{\sinh x} \, \mathrm{d}x \\
&= \int \left(\frac{1}{u} + u \right) \mathrm{d}u \\
&= \ln u + \frac{u^2}{2} + C \\
&= \ln(\sinh x) + \frac{1}{2} \sinh^2 x + C
\end{aligned}$$

Hence

$$\frac{y}{\sinh x} = \ln(\sinh x) + \frac{1}{2} \sinh^2 x + C$$

and so

$$y = \sinh x \ln(\sinh x) + \frac{1}{2} \sinh^3 x + C \sinh x$$

Use the condition $y = 2$ when $x = \ln(1 + \sqrt{2})$. First find $\sinh\big(\ln(1 + \sqrt{2})\big)$:

$$\begin{aligned}
\sinh\big(\ln(1 + \sqrt{2})\big) &= \frac{e^{\ln(1+\sqrt{2})} - e^{-\ln(1+\sqrt{2})}}{2} \\
&= \frac{(1 + \sqrt{2}) - \dfrac{1}{1 + \sqrt{2}}}{2} \\
&= \frac{(1 + \sqrt{2}) - (\sqrt{2} - 1)}{2} \\
&= 1
\end{aligned}$$

So substituting $\sinh x = 1$ and $y = 2$:

$$\begin{aligned}
2 &= 1 \cdot \ln 1 + \frac{1}{2}(1)^3 + C(1) \\
2 &= 0 + \frac{1}{2} + C \\
C &= \frac{3}{2}
\end{aligned}$$

Therefore

$$y = \sinh x \ln(\sinh x) + \frac{1}{2} \sinh^3 x + \frac{3}{2} \sinh x$$

Final answer: $y = \sinh x \ln(\sinh x) + \frac{1}{2} \sinh^3 x + \frac{3}{2} \sinh x$.

14. For $x > 0$, solve the differential equation

$$\sinh x \frac{\mathrm{d}y}{\mathrm{d}x} - y \cosh x = \sinh x \cosh x$$

given that $y = 2$ when $x = \ln(1 + \sqrt{2})$.
Give your answer in exact form. **[7]**

Solution

Since $x > 0$, we have $\sinh x > 0$, so we may divide through by $\sinh x$ to put the differential equation into linear form:

$$\begin{aligned}
\sinh x \frac{\mathrm{d}y}{\mathrm{d}x} - y \cosh x &= \sinh x \cosh x \\
\frac{\mathrm{d}y}{\mathrm{d}x} - \frac{\cosh x}{\sinh x} y &= \cosh x \\
\frac{\mathrm{d}y}{\mathrm{d}x} - \coth x \, y &= \cosh x
\end{aligned}$$

This is a first-order linear equation of the form

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$$

with

$$P(x) = -\coth x$$

So the integrating factor is

$$\begin{aligned}
\mu(x) &= e^{\int -\coth x \, \mathrm{d}x} \\
&= e^{-\int \coth x \, \mathrm{d}x} \\
&= e^{-\ln(\sinh x)} \\
&= \frac{1}{\sinh x}
\end{aligned}$$

Multiplying the differential equation by this integrating factor gives

$$\frac{1}{\sinh x} \frac{\mathrm{d}y}{\mathrm{d}x} - \frac{\coth x}{\sinh x} y = \frac{\cosh x}{\sinh x}$$

The left-hand side is the derivative of $\dfrac{y}{\sinh x}$, so

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{y}{\sinh x}\right) = \coth x$$

Integrating both sides with respect to x:

$$\begin{aligned}
\frac{y}{\sinh x} &= \int \coth x \, \mathrm{d}x + C \\
&= \ln(\sinh x) + C
\end{aligned}$$

Hence

$$y = \sinh x \big(\ln(\sinh x) + C\big)$$

Now use the condition $y = 2$ when $x = \ln(1 + \sqrt{2})$.
We need $\sinh\big(\ln(1 + \sqrt{2})\big)$. Using

$$\sinh u = \frac{e^u - e^{-u}}{2}$$

with $u = \ln(1+\sqrt{2})$:

$$\sinh\big(\ln(1+\sqrt{2})\big) = \frac{(1+\sqrt{2}) - \dfrac{1}{1+\sqrt{2}}}{2}$$

Now rationalise:

$$\begin{aligned}\frac{1}{1+\sqrt{2}} &= \frac{1-\sqrt{2}}{(1+\sqrt{2})(1-\sqrt{2})}\\ &= \frac{1-\sqrt{2}}{1-2}\\ &= \sqrt{2}-1\end{aligned}$$

So

$$\begin{aligned}\sinh\big(\ln(1+\sqrt{2})\big) &= \frac{(1+\sqrt{2}) - (\sqrt{2}-1)}{2}\\ &= \frac{2}{2}\\ &= 1\end{aligned}$$

Substituting into the general solution:

$$\begin{aligned}2 &= 1\big(\ln 1 + C\big)\\ 2 &= 0 + C\\ C &= 2\end{aligned}$$

Therefore the exact solution is

$$y = \sinh x\big(\ln(\sinh x) + 2\big)$$

So the required solution is $y = \sinh x\big(\ln(\sinh x) + 2\big)$

15. Find the general solution of the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} - y\tan x = \frac{1}{\sqrt{1+\sin x}}$$

where $-\frac{\pi}{2} < x < \frac{\pi}{2}$ **[7]**

Solution

Write the differential equation in linear form:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = Q(x)$$

with

$$P(x) = -\tan x, \qquad Q(x) = \frac{1}{\sqrt{1+\sin x}}$$

So the integrating factor is

$$\begin{aligned}\mu(x) &= e^{\int -\tan x\,\mathrm{d}x} \\ &= e^{\ln(\cos x)}\end{aligned}$$

Since $-\frac{\pi}{2} < x < \frac{\pi}{2}$, we have $\cos x > 0$, so

$$\mu(x) = \cos x$$

Multiply the differential equation throughout by $\cos x$:

$$\begin{aligned}\cos x\frac{\mathrm{d}y}{\mathrm{d}x} - y\cos x\tan x &= \frac{\cos x}{\sqrt{1+\sin x}} \\ \cos x\frac{\mathrm{d}y}{\mathrm{d}x} - y\sin x &= \frac{\cos x}{\sqrt{1+\sin x}}\end{aligned}$$

Now

$$\frac{\mathrm{d}}{\mathrm{d}x}(y\cos x) = \cos x\frac{\mathrm{d}y}{\mathrm{d}x} - y\sin x$$

So the equation becomes

$$\frac{\mathrm{d}}{\mathrm{d}x}(y\cos x) = \frac{\cos x}{\sqrt{1+\sin x}}$$

Integrate both sides with respect to x:

$$y\cos x = \int \frac{\cos x}{\sqrt{1+\sin x}}\,\mathrm{d}x + C$$

Let

$$u = 1+\sin x \qquad \Rightarrow \qquad \frac{\mathrm{d}u}{\mathrm{d}x} = \cos x \qquad \Rightarrow \qquad \mathrm{d}u = \cos x\,\mathrm{d}x$$

Then

$$\begin{aligned}\int \frac{\cos x}{\sqrt{1+\sin x}}\,\mathrm{d}x &= \int u^{-1/2}\,\mathrm{d}u \\ &= 2u^{1/2} \\ &= 2\sqrt{1+\sin x}\end{aligned}$$

Hence

$$y\cos x = 2\sqrt{1+\sin x} + C$$

Therefore

$$y = \frac{2\sqrt{1+\sin x}+C}{\cos x}$$

So the general solution is

$$y = \frac{2\sqrt{1+\sin x}+C}{\cos x}$$

An equivalent form is

$$y = \frac{2}{\sqrt{1-\sin x}} + C\sec x$$